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SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0720

COURSE OUTCOME COVERED

CO5: Measure network performance using key metrics with simulation tools.
(BL4,BL5)

Assessment Tool Weightage: 30%

87.1%

Annexure A

Pseudocode Writing Task (Algorithm Design)

1. In a network simulation using Cisco Packet Tracer, a student records total data received as 8000 bits in 4 seconds and observes that 200 packets were sent while 180 packets were received. Write a pseudocode algorithm to calculate the network throughput in bits per second and the packet loss percentage based on the given values. The algorithm should include validation to ensure that time and total packets sent are not zero to avoid errors. It should display both the throughput and packet loss results clearly. Add logic to classify the network as Efficient if throughput is greater than 1500 bps and packet loss is less than 10%, otherwise classify it as Inefficient. Finally, ensure the algorithm outputs a complete summary of the network performance based on the calculated metrics.
2. An organization has multiple network links connecting its branches. The administrator wants to monitor bandwidth utilization and identify overloaded links. Write a pseudocode algorithm that periodically collects the bandwidth usage of each network link and stores the values in an array or list. Use a loop to continuously monitor all links. Add conditions to

Pseudocode Writing Task (Algorithm Design)

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ANSWER:

Calculate Throughput and Packet Loss

Pseudocode

```
BEGIN

    data = 8000
    time = 4
    packets_sent = 200
    packets_received = 180

    IF time = 0 THEN
        PRINT "Invalid time"
        STOP
    END IF

    IF packets_sent = 0 THEN
        PRINT "Invalid packet count"
        STOP
    END IF

    throughput = data / time

    lost_packets = packets_sent - packets_received

    packet_loss = (lost_packets / packets_sent) * 100

    PRINT "Throughput = ", throughput, " bps"
    PRINT "Packet Loss = ", packet_loss, "%"

    IF throughput > 1500 AND packet_loss < 10 THEN
        PRINT "Network is EFFICIENT"
    ELSE
        PRINT "Network is INEFFICIENT"
    END IF

    PRINT "Network Performance Summary"
    PRINT "Data Received = ", data, " bits"
    PRINT "Packets Sent = ", packets_sent
    PRINT "Packets Received = ", packets_received
    PRINT "Packets Lost = ", lost_packets

END
```

Result

Throughput = 2000 bps
Packet Loss = 10%
Network is INEFFICIENT

The network is classified as **Inefficient** because packet loss is 10%, which is not less than 10%.

2. An organization has multiple network links connecting its branches. The administrator wants to monitor bandwidth utilization and identify overloaded links. Write a pseudocode algorithm that periodically collects the bandwidth usage of each network link and stores the values in an array or list. Use a loop to continuously monitor all links. Add conditions to detect when bandwidth utilization exceeds 80% and recommend switching traffic to a backup link. Explain how this algorithm supports efficient bandwidth management and prevents network congestion.

ANSWER:

Monitor Bandwidth Utilization

Pseudocode

```
BEGIN

    links = [Link1, Link2, Link3, Link4]

    WHILE TRUE DO

        FOR each link in links DO

            usage = GET_BANDWIDTH_USAGE(link)

            PRINT link, " Usage = ", usage, "%"

            IF usage > 80 THEN

                PRINT "Warning: Link is overloaded"
                PRINT "Switch traffic to backup link"

            ELSE

                PRINT "Link is working normally"

            END IF

        END FOR

        WAIT 10 seconds

    END WHILE

END
```

Explanation

The algorithm continuously checks the bandwidth usage of each link. If the usage is **more than 80%**, the link is considered overloaded and traffic can be moved to a backup link.

Result

This method helps prevent network congestion, improves bandwidth usage, and maintains smooth communication.

3. A company has six branch offices connected through multiple network links. Each link has different bandwidth, delay, and utilization values, which change throughout the day. The network administrator wants to continuously identify the best communication path between the headquarters and each branch based on overall link quality rather than just the shortest distance. Write a pseudocode algorithm that periodically collects the bandwidth, delay, and utilization of every network link and stores the values in suitable arrays or lists. Use a loop to repeat the monitoring process at regular intervals. Calculate a link quality score for each path using the collected metrics and select the path with the highest score. Include conditions to detect when a link becomes overloaded or fails, automatically selecting an alternate path and generating an alert for the administrator. Explain how your algorithm improves network reliability, routing efficiency, and fault tolerance while adapting to changing network conditions.

ANSWER:

Select the Best Communication Path

Pseudocode

```
BEGIN

    branches = [Branch1, Branch2, Branch3, Branch4, Branch5, Branch6]

    WHILE TRUE DO

        FOR each link DO

            bandwidth = GET_BANDWIDTH(link)
            delay = GET_DELAY(link)
            utilization = GET_UTILIZATION(link)

            IF link is failed THEN

                PRINT "Link failed"
                quality = 0

            ELSE IF utilization > 80 THEN

                PRINT "Link is overloaded"
                quality = 0

            ELSE

                quality = bandwidth - delay - utilization

            END IF

        END FOR

        FOR each branch DO
```

```
        FIND all available paths

        SELECT path with highest quality score

        IF no path is available THEN

            PRINT "No path available"
            SEND ALERT

        ELSE

            PRINT "Best path selected"

        END IF

    END FOR

    WAIT 30 seconds

END WHILE

END
```

Simple Explanation

The algorithm checks three factors:

- **Bandwidth:** Higher is better.
- **Delay:** Lower is better.
- **Utilization:** Lower is better.

The path with the **highest quality score** is selected. If a link fails or becomes more than 80% utilized, it is avoided and an alternative path is selected.

Result

The algorithm provides **better routing, reliability, and fault tolerance** by continuously monitoring network conditions and selecting the best available path.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
A Problem Understanding	20% 2	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
B Algorithm Design	30% 3	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach

C Use of Control Structures	20% 2	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
D Pseudocode Structure	10% 1	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
E Completeness	20% 2	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

	A	B	C	D	E	
Q ₁	2	2	2	1	2	9
Q ₂	2	3	1	1	1	8
Q ₃	1	3	2	1	2	9

26
30

87%